

WASP-43b

R-0482

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230505 · created 2026-07-20T06:47:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460069.6828 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.031
Transit depth (Rp/Rs)^2	3.73 +/- 1.19 [%]
Orbital Inclination (inc)	80.18 +/- 2.18
Airmass coefficient 1 (a1)	1.034 +/- 0.019
Airmass coefficient 2 (a2)	-0.024 +/- 0.047
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#2 - [125, 378]
Optimal Method	PSF photometry
Transit Duration (day)	0.044 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.193 ± 0.031 vs 0.1594 (deviation 1.08σ)
Duration fit vs published	0.044 d vs 0.0512 d (deviation 0.66σ)
Mid-time O–C	2.7 min
Depth SNR	6.2
Residual scatter	1.89 %

Light curve

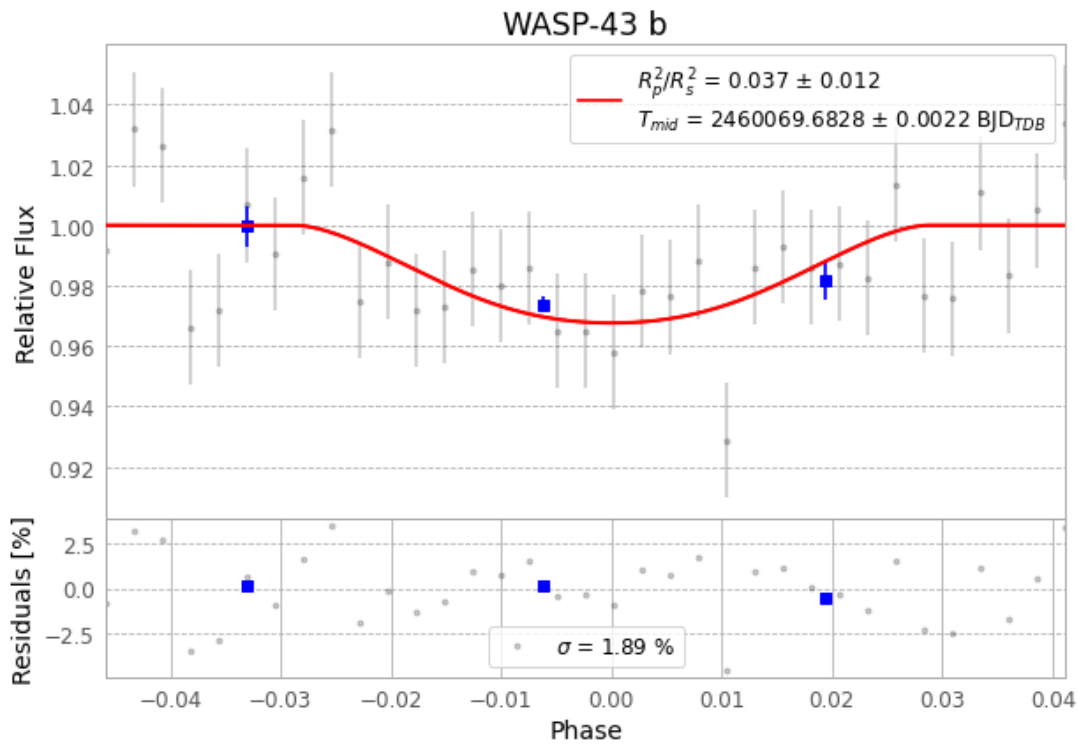


Figure 1 — FinalLightCurve_WASP-43 b_2023-05-04.png (SHA-256 2106198efc049c89...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 333], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a9430798f44465e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

36 FITS frames (24 MB), WASP-43230505032416.FITS → WASP-43230505050916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230505.log (a32054dd96b8...), FinalLightCurve_WASP-43 b_2023-05-04.png (2106198efc04...), FinalLightCurve_WASP-43 b_2023-05-04.csv (5a008afeacd9...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 06:47Z.